

Windows Security -- Guided Notes ANSWER KEY

Unit 3.2 | CompTIA Network+ N10-009 Obj. Core 2 2.2 | N10-008

For instructor use / distribute after students complete the notes.

Question 1

Windows _____ is the built-in antivirus and antimalware solution included with Windows 10 and 11. It provides _____-time protection by scanning files as they are opened or downloaded. When a third-party antivirus is installed, Windows Defender automatically _____.

Answer: Defender (Windows Defender Antivirus); real; disables itself (turns off).

Question 2

Windows _____ is a host-based firewall that blocks _____ connections by default. It maintains separate profiles for _____, _____, and _____ networks. Outbound connections are _____ by default, meaning programs can send data out without a firewall rule.

Answer: Firewall; inbound (unsolicited inbound); Domain; Private; Public; allowed.

Question 3

BitLocker encrypts the entire _____ drive to protect data if the device is lost or stolen. It requires a _____ chip on the motherboard to store the encryption key securely. If the TPM fails or the drive is moved to a different computer, a _____ key (a 48-digit numeric code) is required to unlock the drive.

Answer: System (OS / hard); TPM (Trusted Platform Module); recovery.

Question 4

Windows _____ delivers security patches and bug fixes to the operating system. Delaying or disabling updates leaves the system vulnerable to _____ that target known vulnerabilities. Zero-day vulnerabilities are exploits that exist before a _____ is available.

Answer: Update; exploits (malware / attacks); patch.

Question 5

UAC has four configurable levels. At the highest level, Windows notifies about _____ system changes. At the default level, Windows notifies only when _____ make changes. At a lower level, UAC notifies only when apps try to install software. Disabling UAC entirely removes all _____ before system changes.

Answer: All; apps (programs); prompts (elevation prompts).

Question 6

Safe Mode loads Windows with _____ drivers and services, making it useful for diagnosing and removing malware. Safe Mode with _____ allows internet access for downloading tools. Safe Mode can be accessed by holding _____ while clicking Restart, or by pressing F8 during older system boot sequences.

Answer: Minimal (basic); Networking; Shift.

Question 7

Match each malware type to its description:

Ransomware: _____

Adware: _____

Trojan: _____

Rootkit: _____

Answer: Ransomware: encrypts files and demands payment for decryption. Adware: displays unwanted pop-up or injected ads. Trojan: disguises itself as legitimate software to gain access. Rootkit: hides itself and other malware deep in the OS, often at the kernel level.

Question 8 -- Real World Application

REAL WORLD: A user reports that all their document files now have a .locked extension and cannot be opened. A message on the desktop demands payment in cryptocurrency to restore the files. Describe what type of attack this is, explain why restoring from a backup is the safest response, and list two preventive measures that could have prevented this attack.

Answer: This is a ransomware attack. The malware encrypted the user's files using asymmetric encryption -- decrypting without the attacker's key is computationally infeasible. Paying the ransom does not guarantee key delivery. The safest response is to isolate the machine (disconnect from network), wipe the drive, reinstall Windows, and restore files from an offline or cloud backup taken before the infection. Prevention 1: maintain a 3-2-1 backup (3 copies, 2 media types, 1 offsite or offline) so files can be restored without paying. Prevention 2: keep Windows Update current -- most ransomware exploits known vulnerabilities that have existing patches.